

1. Simulate 30 rolls with `=RANDBETWEEN(1,6)`. What is the probability of rolling a 3 exactly 5 times? (Hint: Use `BINOM.DIST`)
→ The question asks for the probability of rolling a 3 exactly 5 times in 30 rolls of a die.
 - **Number of trials (n):** 30.
 - **Number of successes (x):** 5.
 - **Probability of success (p):** 1/6 (since a die has six sides and we want a specific number).
 - **Excel Function:** `BINOM.DIST(number_s, trials, probability_s, cumulative)`.

The Calculation

In Excel, you would enter:

`=BINOM.DIST(5, 30, 1/6, FALSE)`

Result:

The probability is approximately **0.1921** or **19.21%**. We use `FALSE` because we want the exact probability for 5 successes, not the cumulative total.

2. Generate 100 values in Excel using the continuous uniform distribution `RAND()` and plot a histogram. Describe the shape of the distribution.
→ **Excel Steps**
 1. **Generate Data:** In cell A1, enter `=RAND()` and drag it down to A100. This generates 100 values between 0 and 1.
 2. **Create Plot:** Select the data, go to the **Insert** tab, and select the **Histogram** chart icon.

Shape of the Distribution

- **Appearance:** The histogram will look relatively **flat** or **rectangular**.
 - **Reasoning:** Because it is a **Uniform Distribution**, every value between 0 and 1 has an equal probability of being selected. While there will be some slight "jitter" or unevenness due to the small sample size (100 values), the bars will be roughly the same height across the entire range.
3. A dataset has a mean of 50 and a standard deviation of 5. What percentage of values lie between 45 and 55 if the data follows a normal distribution?

- This question uses the properties of a **Normal Distribution** to find the percentage of data within a specific range.

The Data:

- **Mean (μ):** 50
- **Standard Deviation (σ):** 5
- **Target Range:** 45 to 55

The Calculation

1. **Identify the Standard Deviations:** 45 is exactly one standard deviation below the mean ($50 - 5$).
 - 55 is exactly one standard deviation above the mean ($50 + 5$).
2. **Apply the Empirical Rule:** In a normal distribution, approximately **68%** of the values lie within one standard deviation ($\pm 1\sigma$) of the mean.

Final Answer: Approximately **68%** of the values lie between 45 and 55.

4. What is the concept of standardization (z-score), and why is it important in data analysis? Explain the formula and how standardization transforms a dataset.

- Standardization is the process of putting different datasets on the same scale to compare them fairly.

The Concept

A **z-score** represents how many standard deviations a data point is from the mean. By converting raw data into z-scores, you transform the dataset into a **Standard Normal Distribution**, which always has a mean of 0 and a standard deviation of 1.

The Formula

$$z = (x - \mu) / \sigma$$

- **x:** The raw value.
- **μ :** The population mean.
- **σ :** The population standard deviation.

Why is it Important?

- a. **Comparability:** It allows you to compare values from different scales (e.g., comparing a student's Math score to their Biology score).

- b. **Outlier Detection:** It helps identify outliers; typically, any z-score greater than 3 or less than -3 is considered an outlier.
- c. **Algorithm Requirement:** Many machine learning models (like K-Nearest Neighbors) require standardized data to perform accurately.

5. What is Kurtosis and their type?

→ **Kurtosis** is a statistical measure that describes the "tailedness" of a distribution. It indicates how much of the data is concentrated in the tails versus the center compared to a normal distribution.

There are three main types of Kurtosis:

- a. **Mesokurtic:** This describes a **Normal Distribution**. The tails are neither too thick nor too thin, and the distribution has a kurtosis value of approximately 3 (or an "excess kurtosis" of 0).
- b. **Leptokurtic:** This distribution has **fat tails** and a **sharp, tall peak**. It indicates a high concentration of data around the mean and a higher likelihood of extreme outliers. (Excess kurtosis > 0).
- c. **Platykurtic:** This distribution is **flat** with **thin tails**. The data is spread out more widely, resulting in a shorter peak compared to a normal distribution. (Excess kurtosis < 0).

6. Explain why the uniform distribution is a good model for the outcome of rolling a fair die

→ A **Discrete Uniform Distribution** is a model where all possible outcomes are equally likely to occur.

Why it fits a fair die:

- a. **Equal Probability:** A fair die has 6 faces, and each face ($\{1, 2, 3, 4, 5, 6\}$) has exactly a $1/6$ chance of landing face up.
- b. **Finite Outcomes:** There is a fixed, countable number of results, which is the definition of a discrete distribution.
- c. **Consistency:** Because the die is "fair," no single number is weighted to appear more often than others, resulting in a "flat" probability distribution where $P(X=1) = P(X=2) = \dots = P(X=6)$.

7. Use Excel to compute the probability of getting at least 8 successes in 15 trials with success probability 0.5

→ A question asks for "at least" a certain number, it means that specific number or any number higher. In this case, we want the probability of getting 8, 9, 10, 11, 12, 13, 14, or 15 successes.

The Data:

- **Trials (n):** 15
- **Success Probability (p):** 0.5
- **Target:** ≥ 8 successes

The Excel Logic

Excel's `BINOM.DIST` function with the `TRUE` argument calculates **cumulative** probability (from 0 up to x). Since we want the "upper tail" (8 and above), we subtract the "lower tail" (0 through 7) from the total probability of 1.

The Formula:

`=1 - BINOM.DIST(7, 15, 0.5, TRUE)`

Result:

The probability is **0.5** or **50%**.

Note: Because the success probability is exactly 0.5, the distribution is perfectly symmetrical. In 15 trials, the median is 7.5, so the probability of being above the median (8+) is equal to the probability of being below it.

8. How does log transformation help in stabilizing variance and making data more normally distributed?

→ Log transformation is a powerful technique used to replace each variable x with $\log(x)$. It is particularly useful when dealing with skewed data or inconsistent variability.

1. Stabilizing Variance

- **The Issue:** In many datasets, as the mean increases, the variance (the spread of the data) also increases. This is known as heteroscedasticity.
- **The Fix:** A log transformation "compresses" the higher values more than the lower values. This shrinking effect helps make the variance constant across the

entire range of the data, which is a requirement for many statistical tests like ANOVA or linear regression.

2. Making Data More Normally Distributed

- **The Issue:** Real-world data (like income or population) is often **right-skewed** (positively skewed), meaning there are a few very large values that create a long tail to the right.
- **The Fix:** By applying a log transformation, the long right tail is pulled back toward the center, while the smaller values are spread out. This often results in a distribution that looks much more like a symmetrical **Bell Curve** (Normal Distribution).

3. Making Relationships Linear

- Log transformations can turn exponential relationships into linear ones, making them much easier to model and interpret using standard regression techniques.